

Bahram Behzadian

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Reinforcement Learning | Robust Decision-Making | Post-Training

Google Scholar | U.S. permanent resident

Research Profile

Research scientist studying policy improvement when the model of the world or the evaluator is imperfect. My work spans robust MDPs under uncertain dynamics and policy improvement under truncated evaluators with verifier-style rewards, with applications to RL post-training and reasoning.

Research Highlights

Policy Improvement under Imperfect Evaluators

UPI-TRM, ICLR 2026 Workshop on AI with Recursive Self-Improvement

Derived finite-depth policy-improvement guarantees that separate Bellman residual from truncation bias, and studied verifier-style rewards in recursive reasoning models.

Robust and Risk-Sensitive Markov Decision Processes

NeurIPS 2021 and AISTATS 2021

Developed scalable algorithms for L_∞ -constrained S-rectangular robust MDPs and percentile criteria under model uncertainty.

Experience

2022–Present **Research Scientist**, *Meta*

- Build production systems for experiment-metric correctness, anomaly and data-quality monitoring, and ML auditing in large-scale ads experimentation.
- Develop statistical diagnostics that identify unreliable, biased, stale, or inconsistent signals and translate failure analyses into monitoring and correctness safeguards.

2018 **AI Engineer Intern**, *Envio, Inc.*, Dover, NH

- Developed learning-based methods for vehicle routing and scheduling under operational constraints and uncertain execution.

Selected Publications

UPI-TRM: Policy Improvement under Truncated Evaluators with Verifier-Style Rewards

Bahram Behzadian, Brett Daley, Gopeshh Subbaraj, Houssam Nassif

ICLR 2026 Workshop on AI with Recursive Self-Improvement [Paper]

Fast Algorithms for L_∞ -Constrained S-Rectangular Robust MDPs

Bahram Behzadian, Marek Petrik, Chin Pang Ho

Conference on Neural Information Processing Systems (NeurIPS), 2021 [Paper]

Optimizing Percentile Criterion using Robust MDPs

Bahram Behzadian, Reazul Hasan Russel, Marek Petrik, Chin Pang Ho

International Conference on Artificial Intelligence and Statistics (AISTATS), 2021 [Paper]

Fast Feature Selection for Linear Value Function Approximation

Bahram Behzadian, Soheil Gharatappeh, Marek Petrik

International Conference on Automated Planning and Scheduling (ICAPS), 2019 [Paper]

Monte Carlo Localization in Hand-Drawn Maps

Bahram Behzadian, Pratik Agarwal, Wolfram Burgard, Gian Diego Tipaldi

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2015 [Paper]

Complete publication list: [Publications page](#) | [Google Scholar](#)

Education

2015–2022 **Ph.D. in Computer Science**, *University of New Hampshire*

Dissertation: *Efficient Data-Driven Robust Policies for Reinforcement Learning* (advisor: Marek Petrik).

M.S. in Computer Science awarded en route (2018). Thesis: *Representation Learning for Reinforcement Learning via Singular Value Decomposition*.

2010–2013 **M.S. in Machine Automation**, *Tampere University*

Thesis: *Robot Localization under Partial Observability with Weak Maps*

Honors and Fellowships

CEPS Graduate Fellowship, University of New Hampshire

Thesis and Dissertation Fellowship, Tampere University of Technology

Technical Skills

Research: robust MDPs, policy improvement, model-based RL, partial observability, verifier-style rewards, RLVR.

Post-training: PPO, DPO, GRPO, reward modeling.

Programming: Python, PyTorch.